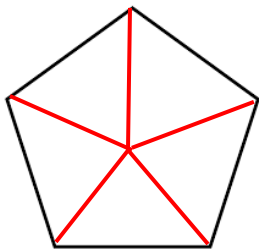


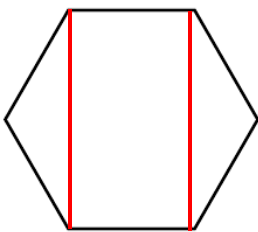
Grade 7
Unit 4
Lesson 3 Practice

Name Answers
Date _____
Period _____

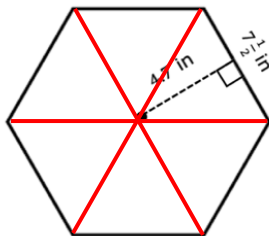
1. Decompose the regular pentagon into only triangles.



2. Decompose the regular hexagon into triangles and quadrilaterals.

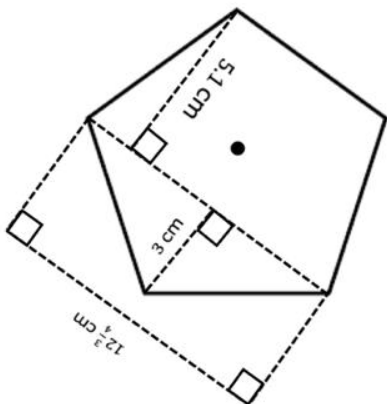


3. A hexagon is given in inches (in). Each side of the hexagon has a length of $7\frac{1}{2}$ inches and a perpendicular distance from the center to the midpoint of a side to be 4.7 inches. Find the area of the regular hexagon rounded to the nearest square inch.



$$A = \left(\frac{1}{2} \cdot 7\frac{1}{2} \cdot 4.7 \right) \cdot 6 \text{ triangles} \\ = 105.75 \text{ in}^2$$

4. Find the area of the regular pentagon that has a side length of 6 cm.

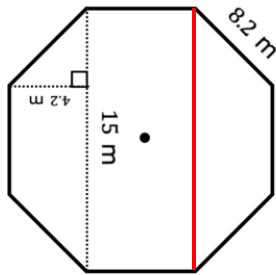


$$\text{Trapezoid: } A = \frac{1}{2} \cdot 5.1 \cdot \left(6 + 12\frac{3}{4} \right) = 47.8125$$

$$\text{Triangle: } A = \frac{1}{2} \cdot 3 \cdot 12.75 = 19.125$$

$$47.8125 + 19.125 = 66.9375 \text{ cm}^2$$

5. An octagon is shown with lengths in meters (m). What is the area of the octagon?



$$\text{Trapezoids} \rightarrow 2 \left(\frac{1}{2} \cdot 4.2 \cdot (8.2 + 15) \right) = 2(48.72) = 97.44$$

$$\text{Rectangle} \rightarrow 15 \cdot 8.2 = 123$$

$$123 + 97.44 = 220.44 \text{ m}^2$$

6. Label the regular decagon with the following dimensions.

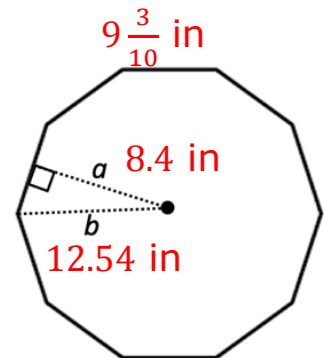
$$\text{Side length} = 9\frac{3}{10} \text{ in.}$$

$$a = 8.4 \text{ in.}$$

$$b = 12.54 \text{ in.}$$

Which of the dimensions is not needed to find the area of the decagon?

b is neither a base nor a height



7. Find the area of the decagon from question 8 in square inches.

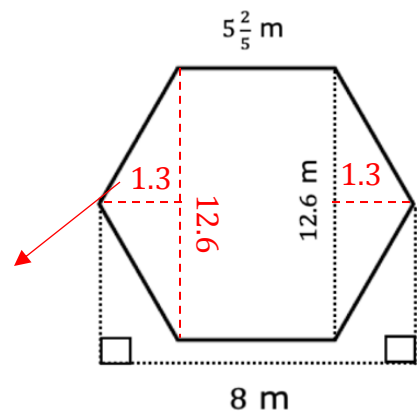
$$A = 10(9.3 \cdot 8.4 \cdot 0.5)$$

$$= 10(39.06)$$

$$= 390.6 \text{ in}^2$$

8. Mia wanted to find the area of the hexagon by decomposing it into two triangles and a quadrilateral. Help her decompose the hexagon and label any additional necessary dimensions.

$$\frac{8 - 5\frac{2}{5}}{2} =$$



9. Find the area of the hexagon above in square meters.

$$\text{Triangles} \rightarrow 2(0.5 \cdot 1.3 \cdot 12.6)$$

$$2(8.19)$$

$$= 16.38 \text{ m}^2$$

$$\text{Rectangle} \rightarrow 5.4 \cdot 12.6$$

$$= 68.04 \text{ m}^2$$

$$16.38 + 68.04 = 84.42 \text{ m}^2$$